

Next Generation Data Centers and Energy Management Systems

Lecture 3A: Performance Metrics and Standards

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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- **Performance Metrics and Standards**
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Define data center benchmark metrics (e.g., PUE).

Typical layout of a data center facility

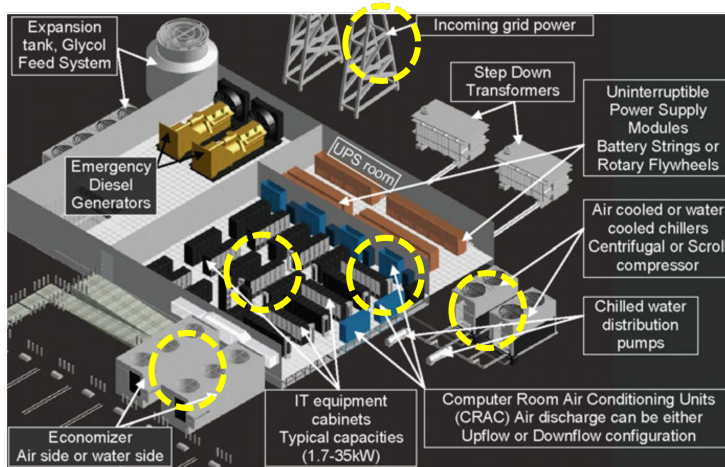


Figure 1: (Figure 1.1 in Ref. [?]).

Simulink and Simscape

We can use different types of software to perform thermofluids flow network analysis. I have a few examples posted in our Github here: [examples](#). These run in Matlab. We can do similar things in Cadence, Modelica, etc.

Historical development of PUE

2007, U.S. Environmental Protection Agency (EPA):

- “The federal government and industry should work together to develop an objective, credible energy performance rating system for data centers, initially addressing the infrastructure portion but extending, when possible, to include a companion metric for the productivity and work output of IT equipment (ITE).”

PUE

PUE: A Comprehensive Examination of the Metric

ASHRAE Standards

- Read Only versions of ASHRAE standards can be accessed here: [link](#).
- See: Standard 90.4-2025 – Energy Standard for Data Centers.
- The purpose of the standard (90.4-2025) is to establish minimum energy efficiency requirements for the design, construction, and planned operation and maintenance of data centers, and to balance those requirements with related factors affecting the environment. Factors to be evaluated include the following:
 - Nonrenewable energy consumption
 - Renewable energy consumption
 - Water consumption
 - Greenhouse gas emissions
- The standard applies to new data centers, new additions to data centers, and modifications of data centers.
- All non-data center spaces in the building must demonstrate compliance with ASHRAE Standard 90.1-2025 (see [COMcheck](#)).

Annualized Mechanical Load Component and Design Electrical Loss Component (Design ELC)

Electrical distribution monitoring

- Facility level (total building kWh, PUE):
 - Smart utility meters measure the total power consumption of the entire facility.
 - Modern switchgears have digital meters to monitor voltage and current.
 - Uninterruptible power supply systems can also be used for monitoring (uptime and efficiency).
- Rack-level Power Monitoring (Watts per rack/outlet):
 - Intelligent power distribution units. Power strips inside of the server racks.
- Sensors (ambient environment):
 - Temperature and humidity sensors, located in the top, middle, and bottom of racks (inlet and outlet).
 - Airflow and differential pressure sensors.
 - Liquid cooling flow meters.
- Software:
 - Data center infrastructure management. Determines PUE in real-time.
 - Building management system, focuses on chillers, CRACs, etc.

Ensemble coefficient of performance

Data center energy use

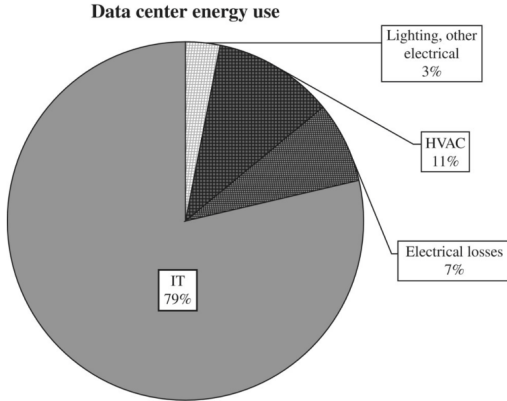


Figure 2: Reproduced from Ref. [?].

- Largest power consumer in the mechanical system is the chiller.
- One strategy to reduce energy use is to elevate the supply air temperature by increasing the chilled water supply temperature.
- This strategy depends on the type of mechanical system, the climate, and the allowable air temperature for the IT equipment.

References I

- [1] Hwaiyu Geng.
Data center handbook.
John Wiley & Sons, 2021.
- [2] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.